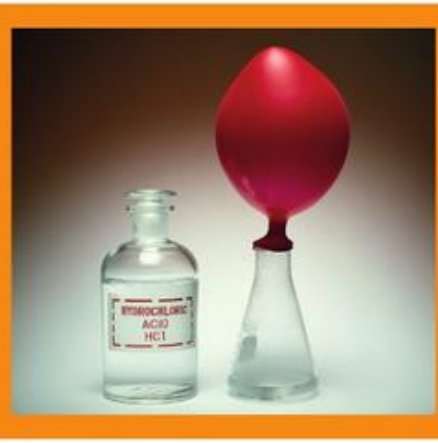
CHEMISTRY ONLY

6 CHEMICAL FORMULAE, EQUATIONS AND CALCULATIONS: PART 2

In this chapter we will look at how to extend our understanding of moles to carry out calculations involving gases and solutions. You need to have studied the material covered in Chapter 5 before starting on this chapter.

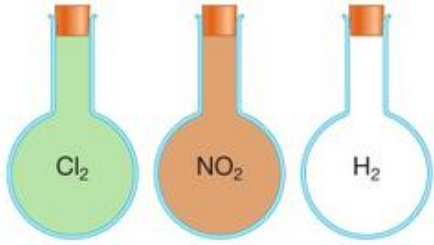


▶ Figure 6.1 When magnesium reacts with hydrochloric acid, hydrogen gas is formed. In this chapter you will learn to calculate what volume of gas is produced.

LEARNING OBJECTIVES

- Understand how to carry out calculations involving amount of substance, volume and concentration (in mol/dm^3) of solution
- Understand how to carry out calculations involving gas volumes and the molar volume of a gas (24 dm^3 and 24 000 cm^3 at room temperature and pressure (rtp))

CALCULATIONS INVOLVING GAS VOLUMES



▲ Figure 6.2 Three identical flasks containing different gases at the same temperature and pressure all contain equal numbers of molecules.

KEY POINT

For helium and the other noble gases we would use the word 'atoms' instead of 'molecules'.

Avogadro's law:

Equal volumes of gases at the same temperature and pressure contain equal numbers of molecules.

This means that if you have 100 cm^3 of hydrogen at some temperature and pressure, it contains exactly the same number of molecules as there are in 100 cm^3 of Cl_2, or any other gas under those conditions, irrespective of the size of the molecules.

You can see how this works in a chemical reaction. Methane (CH_4) reacts with oxygen to form carbon dioxide and water. The equation for the reaction is:

$$CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(l)$$

We need two O_2 molecules to react with each CH_4 molecule, so if we react 100 cm^3 of CH_4 we will need twice the volume of O_2 to obtain the correct number of particles. In this equation, 100 cm^3 of methane will react exactly with 200 cm^3 of oxygen gas and 100 cm^3 of carbon dioxide gas will be formed.

UNITS OF VOLUME

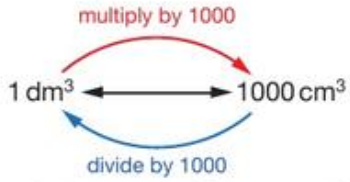
KEY POINT

If you want to talk about 1000 cm^3, the cubic decimetre is the preferred unit rather than the litre.

Volumes (of gases or liquids) are measured in cubic centimetres (cm^3)
or cubic decimetres (dm^3)
or litres (l)

1 litre = 1 dm^3 = 1000 cm^3

The most important thing you need to be able to do is to convert between cm^3 and dm^3. Figure 6.3 shows how you can do this.



▲ Figure 6.3 Because 1 dm^3 is equivalent to 1000 cm^3, to convert dm^3 to cm^3 we have to go from 1 to 1000, therefore we need to multiply by 1000. To convert cm^3 to dm^3 we need to go from 1000 down to 1, therefore we divide by 1000.

THE VOLUME OCCUPIED BY 1 MOLE OF A GAS

KEY POINT

This is by far the most important consequence of Avogadro's law.

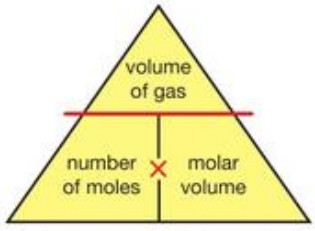
1 mole of any gas contains the same number of molecules and so occupies the same volume as 1 mole of any other gas at the same temperature and pressure.

At room temperature and pressure, the volume occupied by 1 mole of any gas is approximately 24 dm^3 (24 000 cm^3). The volume occupied by 1 mole of a gas is often called the **molar volume**.

KEY POINT

1 mole of any gas occupies 24 dm^3 (24 000 cm^3) at rtp.
The abbreviation **rtp** is commonly used for 'room temperature and pressure'. rtp is usually taken as 20–25 °C and 1 atmosphere pressure.

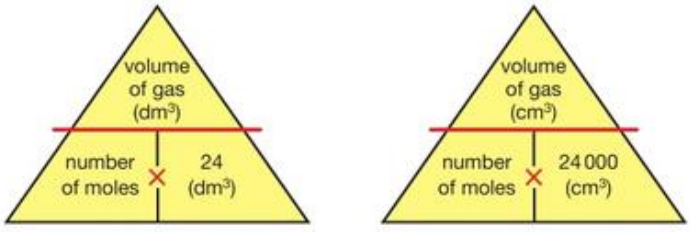
You can use the triangles shown in Figures 6.4 and 6.5 to convert between volumes and numbers of moles.



▲ Figure 6.4 This triangle can be used to convert between volume of a gas and number of moles.

EXTENSION WORK

To be absolutely 100% correct, the molar volume has units of dm^3/mol or cm^3/mol. At International GCSE you are unlikely to encounter this unit.



▲ Figure 6.5 These are basically the same as the triangle in Figure 6.4 except that the molar volume is shown as a number. Be careful with the units of volume.

CALCULATIONS WITH MOLAR VOLUME

CALCULATING THE VOLUME OF A CERTAIN NUMBER OF MOLES

Calculate the volume in dm^3 of 0.20 mol CO_2 at rtp:

volume = number of moles × molar volume

Because we want the volume in dm^3 we use 24 dm^3 as the molar volume:

volume = 0.20 × 24 = 4.8 dm^3

CALCULATING THE VOLUME OF A GIVEN MASS OF GAS

Calculate the volume (in cm^3) of 0.01 g of hydrogen at rtp (A_r: H = 1).

1 mol H_2 has a mass of 2 g:

$$\text{number of moles} = \frac{\text{mass}}{\text{mass of 1 mol}}$$

0.01 g of hydrogen is $\frac{0.01}{2}$ mol = 0.005 mol.

Because we want the volume in cm^3 we use the molar volume as 24 000 cm^3:

volume = number of moles × molar volume

0.005 mol of hydrogen occupies 0.005 × 24 000 = 120 cm^3

CALCULATING THE NUMBER OF MOLES FROM A VOLUME

You have to be careful that the units are correct when you are calculating a number of moles from a volume. Look carefully, is the volume of gas given in cm^3 or dm^3? If the volume of the gas is given in cm^3 then use the molar volume of the gas as 24 000 cm^3, if it is given in dm^3 then use 24 dm^3.

Calculate the amount of moles in 120 cm^3 of carbon dioxide.

We can see that the volume is given in cm^3 therefore we use 24 000 cm^3 as the molar volume:

$$\text{number of moles} = \frac{\text{volume of gas}}{\text{molar volume}}$$

$$\text{number of moles} = \frac{120}{24\,000}$$

number of moles = 0.005 mol

USING THE MOLAR VOLUME IN CALCULATIONS WITH CHEMICAL EQUATIONS

In order to do these calculations, we follow basically the same method as we used for masses in Chapter 5.

- Calculate the number of moles of anything you can.
- Use the chemical equation to deduce the number of moles of what you want.
- Convert the number of moles to the required quantity, e.g. a mass or a volume.

KEY POINT

Excess means that more than enough acid has been added to react with all the calcium carbonate.

Calculate the volume of carbon dioxide produced at room temperature and pressure when an excess of dilute hydrochloric acid is added to 1.00 g of calcium carbonate. (A_r: C = 12, O = 16, Ca = 40; molar volume = 24 dm^3 at rtp.)

The equation for the reaction is

$$CaCO_3(s) + 2HCl(aq) \rightarrow CaCl_2(aq) + CO_2(g) + H_2O(l)$$

We have been given the mass of calcium carbonate ($CaCO_3$) and so we can calculate the number of moles of calcium carbonate:

$$\text{number of moles} = \frac{\text{mass}}{\text{mass of 1 mole}}$$

The mass of 1 mole of $CaCO_3$ is the M_r in grams. The M_r is 40 + 12 + (3 × 16) = 100.

The mass of 1 mole of $CaCO_3$ is 100 g.

The number of moles in 1.00 g of $CaCO_3$ = $\frac{1.00}{100}$ = 0.0100 mol.



▲ Figure 6.6 Shells of sea creatures such as limpets are made of calcium carbonate, $CaCO_3$. If they were 100% pure, 1 g of shells would react with hydrochloric acid to give 0.24 dm^3 of carbon dioxide.

From the chemical equation we can see that 1 mol of $CaCO_3$ produces 1 mol of CO_2 so we know that 0.0100 mol of $CaCO_3$ will produce the same number of moles of CO_2, that is, 0.0100 mol.

We now need to work out the volume occupied by 0.0100 mol of CO_2:

volume of gas = number of moles × molar volume

= 0.0100 × 24 = 0.24 dm^3

We have used the molar volume in dm^3 and so our answer will be in dm^3. It is really important to include the units here because the examiner will not know whether you mean cm^3 or dm^3 unless you write it down.

A PROBLEM INVOLVING MAKING HYDROGEN

Aluminium reacts with dilute hydrochloric acid according to the equation

$$2Al(s) + 6HCl(aq) \rightarrow 2AlCl_3(aq) + 3H_2(g)$$

What mass of aluminium would you need to add to an excess of dilute hydrochloric acid so that you produced 100 cm^3 of hydrogen at room temperature and pressure? (A_r of Al = 27; molar volume = 24 000 cm^3 at rtp.)

We can only calculate the number of moles of the hydrogen gas. We do not know the mass of aluminium, that is what we are trying to find.

The volume of hydrogen gas is given in cm^3 so we use the molar volume of a gas as 24 000 cm^3:

$$\text{number of moles of gas} = \frac{\text{volume of gas}}{\text{molar volume}} = \frac{100}{24\,000} = 0.00417 \text{ mol}$$

We could also write this number in standard form as 4.17×10^{-3} mol.

Looking at the chemical equation we can see that 3 mol of H_2 is produced from 2 mol of Al (we are only looking at the big numbers in front of the formulae). So we can see that the number of moles of Al will be $\frac{2}{3}$ times the number of moles of H_2.

The number of moles of aluminium required to produce 0.00417 mol of H_2 is $0.00417 \times \frac{2}{3} = 0.00278$ mol.

We can convert this to a mass by using the equation:

mass = number of moles × mass of 1 mol

= 0.00278 × 27 = 0.075 g Al

HINT

Another way of thinking about this is that to go from 3 (H_2) to 2 (Al) we divide the 3 by 3 to get 1 and then multiply this by 2 to get 2. In other words, to go from the number of moles of H_2 to the number of moles of Al, we divide the number of moles of H_2 by 3 and then multiply by 2.

REMINDER

The equation to calculate the mass is the number of moles × mass of 1 mol. It is important to multiply by 27 here and not 54. Although it is 2Al in the equation, we have already used the 2 in calculating the number of moles of Al. Also, the mass of 0.00278 mol of Al must always be the same, it cannot depend on what it is reacting with. 0.00278 mol of Al always contains the same number of atoms.

WORKING WITH SOLUTION CONCENTRATIONS

CONCENTRATIONS OF SOLUTIONS

Concentrations can be measured in either

- g/dm^3

or

- mol/dm^3.

These can also be written as g dm^{-3} and mol dm^{-3}. You read them as 'grams per cubic decimetre' and 'moles per cubic decimetre'. Remember: 1 cubic decimetre is the same as 1 litre.

You have to be able to convert between g/dm^3 and mol/dm^3. This is no different from converting moles into grams and *vice versa*. When you are doing this

KEY POINT

You may also find the symbol M used. For example, dilute hydrochloric acid might have a concentration quoted as 2 M. M means 'mol/dm^3' and is described as the **molarity** of the solution. You can also read 2 M as '2 molar'.



▲ Figure 6.7 Sea water contains about 0.6 moles of NaCl per cubic decimetre.

conversion in concentration calculations, it does not make any difference that the amount of substance you are talking about is dissolved in 1 dm^3 of solution.

A sample of sea water has a concentration of sodium chloride of 35.1 g/dm^3. Find its concentration in mol/dm^3. (A_r: Na = 23, Cl = 35.5)

1 mol NaCl has a mass of 58.5 g.

There are 35.1 g in every dm^3 of solution

$$\text{number of moles} = \frac{\text{mass(g)}}{\text{mass of 1 mole(g)}}$$

$$35.1\,\text{g is } \frac{35.1}{58.5}\,\text{mol} = 0.600\,\text{mol}$$

Therefore there are 0.600 mol in every dm^3 of solution and the concentration of the NaCl is 0.600 mol/dm^3.

EXAMPLE 1

What is the concentration of a 0.050 mol/dm^3 solution of sodium carbonate, Na_2CO_3, in g/dm^3? (A_r: C = 12, O = 16, Na = 23)

1 dm^3 of solution contains 0.050 mol Na_2CO_3.

1 mol Na_2CO_3 weighs 106 g.

0.050 mol weighs $0.050 \times 106 = 5.3$ g.

1 dm^3 of solution contains 5.3 g Na_2CO_3, therefore the concentration is 5.3 g/dm^3.

EXAMPLE 2

What is the concentration in mol/dm^3 of a solution containing 2.1 g of sodium hydrogencarbonate, $NaHCO_3$, in 250 cm^3 of solution? (A_r: H = 1, C = 12, O = 16, Na = 23)

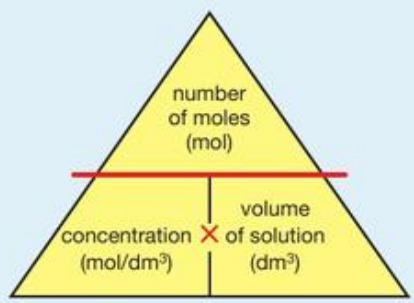
1 mol $NaHCO_3$ has a mass of 84 g.

$$2.1\,\text{g is } \frac{2.1}{84} = 0.025\,\text{mol}$$

This is in a volume of 250 cm^3 but we need to find out how much there is in 1 dm^3 (1000 cm^3). There are four lots of 250 cm^3 in 1000 cm^3. Each portion of 250 cm^3 contains 0.025 mol, therefore there must be $4 \times 0.025 = 0.10$ mol in 1000 cm^3.

The concentration is 0.10 mol/dm^3.

Another way of doing this is to use the triangle shown in Figure 6.8.



▲ Figure 6.8 We can use the triangle for calculations involving solutions.

WORKING OUT A NUMBER OF MOLES FROM A VOLUME AND A CONCENTRATION

When doing calculations involving solutions you will most often be asked to calculate a number of moles given the volume of the solution and its concentration. The equation for doing this is:

number of moles = volume of solution (dm^3) × concentration (mol/dm^3)

Because concentration is usually given in units of mol/dm^3 but volumes of solutions are often given in cm^3, we usually have to convert the volume in cm^3 to dm^3 by dividing by 1000.

EXAMPLE 3

Calculate the number of moles of NaOH in 50 cm^3 of 0.10 mol/dm^3 solution.

Converting the volume to dm^3 we get 50/1000 = 0.050 dm^3.

Number of moles = volume of solution (dm^3) × concentration (mol/dm^3).

Number of moles = 0.050 × 0.10 = 0.0050 mol.

CALCULATIONS WITH EQUATIONS INVOLVING SOLUTIONS

A CALCULATION INVOLVING HARD WATER

Limescale can be removed from, for example, electric kettles by reacting it with a dilute acid such as ethanoic acid, which is present in vinegar:

$$CaCO_3(s) + 2CH_3COOH(aq) \rightarrow (CH_3COO)_2Ca(aq) + CO_2(g) + H_2O(l)$$

HINT

Don't be scared that some of this is new to you! As long as you realise that CH_3COOH is ethanoic acid, that's all you need to worry about for this calculation.

What mass of calcium carbonate can be removed by 50 cm^3 of a solution of ethanoic acid that has a concentration of 2 mol/dm^3?
(A_r: C = 12, O = 16, Ca = 40)

Again, we will use our three stages:

- calculate the number of moles of what we can
- use the chemical equation to work out the number of moles of what we want
- convert the number of moles to what is required in the question.

We know the volume and concentration of ethanoic acid and so we can calculate the number of moles of ethanoic acid:

number of moles = volume of solution (dm^3) × concentration (mol/dm^3)

We need to be careful to convert the volume of ethanoic acid to dm^3 by dividing by 1000:

$$\text{number of moles of ethanoic acid} = \frac{50}{1000} \times 2 = 0.1 \text{ mol}$$

From the chemical equation we can see that there is a 2 in front of the CH_3COOH (ethanoic acid) but no number (which means a 1) in front of the $CaCO_3$, so we can deduce that 2 mol of CH_3COOH react with 1 mol of $CaCO_3$, in other words the number of moles of $CaCO_3$ is half the number of moles of CH_3COOH. Therefore, we can deduce that 0.1 mol of ethanoic acid will react with 0.1/2 = 0.05 mol of $CaCO_3$.

We can now calculate the mass of $CaCO_3$ (M_r = 100):

mass = number of moles × mass of 1 mol = 0.05 × 100 = 5 g

Therefore the ethanoic acid reacts with 5 g of calcium carbonate.

CALCULATIONS FROM TITRATIONS

ACID–ALKALI TITRATIONS

HINT

The acid and alkali can also be added the other way round, with the alkali in the burette.

Titration is a technique that is used to find out how much of one solution reacts with a certain volume of another solution of known concentration. A solution of an **alkali** is measured into a conical flask using a pipette. An **acid** is run in from a burette, swirling the flask constantly. Towards the end, the acid is run in a drop at a time until the **indicator** just changes colour. If you know the concentration of either the acid or the alkali you can use the results of the titration to find the concentration of the solution you reacted it with.



▲ Figure 6.9 The endpoint of a titration using methyl orange as indicator.

Acid–alkali titrations are discussed in Chapters 16 and 17.

THE STANDARD CALCULATION

A standard titration problem will look like this:

25.00 cm^3 of 0.100 mol/dm^3 sodium hydroxide solution required 23.50 cm^3 of dilute hydrochloric acid for **neutralisation**. Calculate the concentration of the hydrochloric acid.

$$NaOH(aq) + HCl(aq) \rightarrow NaCl(aq) + H_2O(l)$$

We will use the same three-step method as we used above to calculate the concentration of the hydrochloric acid.

HINT

Remember to divide the volume by 1000 to convert to dm^3.

We know the volume (25.00 cm^3) and concentration (0.100 mol/dm^3) of the sodium hydroxide solution and so we can calculate the number of moles:

$$\text{number of moles of NaOH} = \frac{25.00}{1000} \times 0.100 = 0.00250 \text{ mol}$$

From the chemical equation we know that 1 mol of NaOH reacts with 1 mol of HCl.

Therefore 0.00250 mol NaOH reacts with 0.00250 mol HCl.

Now that we have the number of moles of HCl we need to convert it to a concentration.

We know that we need 0.00250 mol of HCl to neutralise the NaOH. The volume of HCl we added was 23.50 cm^3. So we know that this volume must contain 0.00250 mol of HCl.

We have the volume and number of moles of HCl and can work out the concentration:

$$\text{concentration } (mol/dm^3) = \frac{\text{number of moles (mol)}}{\text{volume } (dm^3)}$$

Because we want to calculate concentration in mol/dm^3 we have to convert the volume of HCl to dm^3 by dividing by 1000:

$$\text{volume of HCl} = \frac{23.50}{1000} = 0.02350\,\text{dm}^3$$

$$\text{concentration} = \frac{0.00250}{0.02350} = 0.106\,\text{mol/dm}^3$$

The concentration of the hydrochloric acid is 0.106 mol/dm^3.

A VERY SLIGHTLY HARDER CALCULATION

25.0 cm^3 of sodium hydroxide solution of unknown concentration was titrated with dilute sulfuric acid of concentration 0.050 mol/dm^3. 20.0 cm^3 of the acid was required to neutralise the alkali. Find the concentration of the sodium hydroxide solution in mol/dm^3.

$$2NaOH(aq) + H_2SO_4(aq) \rightarrow Na_2SO_4(aq) + 2H_2O(l)$$

This time, we know everything we need about the sulfuric acid and can calculate the number of moles.

The experiment used 20.0 cm^3 of 0.050 mol/dm^3 H_2SO_4:

$$\text{number of moles of sulfuric acid used} = \frac{20.0}{1000} \times 0.050$$

$$= 0.0010\,\text{mol}$$

The equation proportions aren't 1:1 this time. That's what makes the calculation slightly different from the last one. The equation shows that 1 mol of sulfuric acid reacts with 2 mol of sodium hydroxide. So the number of moles of sodium hydroxide is twice the number of moles of sulfuric acid:

$$\text{number of moles of sodium hydroxide} = 2 \times 0.0010 = 0.0020\,\text{mol}$$

That 0.0020 mol must have been in the 25.0 cm^3 (25/1000 = 0.025 dm^3) of sodium hydroxide solution:

$$\text{concentration (mol/dm}^3\text{)} = \frac{\text{number of moles (mol)}}{\text{volume (dm}^3\text{)}}$$

$$\text{concentration} = 0.0020/0.025 = 0.080\,\text{mol/dm}^3$$

Therefore the concentration of the sodium hydroxide is 0.080 mol/dm^3.

REVERSING THE CALCULATIONS

Instead of calculating the concentration of a solution using titration results, you may be asked to calculate what volume of a solution is needed to neutralise something else. Here is an example.

EXAMPLE 4

Calculate the volume of 0.100 mol/dm^3 sodium hydrogencarbonate ($NaHCO_3$) solution needed to neutralise 20.0 cm^3 of 0.125 mol/dm^3 hydrochloric acid (HCl).

$$NaHCO_3(aq) + HCl(aq) \rightarrow NaCl(aq) + CO_2(g) + H_2O(l)$$

We have been given the volume and concentration of the hydrochloric acid and so we can calculate the number of moles of this. We do not have enough information to calcuate the number of moles of anything else, so must start here:

$$\text{number of moles of HCl} = \frac{20.0}{1000} \times 0.125 = 0.00250\,\text{mol}$$